A Hierarchical Region-Merging Algorithm for 3-D Segmentation of Individual Trees Using UAV-LiDAR Point Clouds

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***Abstract*— Over an extended period, remote-sensing-based individual tree analysis has played a critical role in modern forest inventory and management research. The segmentation of individual trees from aerial point clouds usually depends on the characteristics of peak-like uplift on the crown surface; however, the performance inevitably decreases with increasing visibility of such features in point clouds, especially for high-density forests. Herein, we developed a novel hierarchical region-merging algorithm that first over-segmented the entire forest scene based on local density and then merged the over-segmented partitions into pairs through a stepwise optimal process to produce the final segmentation. In the region-merging method, a global merging cost was introduced to shift from local detection of crown features to use the overall compactness of forest point clouds. The experiments were conducted using unmanned aerial vehicle light detection and ranging (UAV-LiDAR) point clouds from three coniferous stands with different densities and a high-density coniferous and broad-leaved mixed stand. A total of 5510 field-measured trees in 36 plots were used to assess the accuracy of the proposed method. Our method achieved F-scores of 0.91, 0.88, 0.84, and 0.80 for low- (∼700 stems/ha), medium- (∼1000 stems/ha), and high-density (∼2000 stems/ha) conifer stands and coniferous and broad-leaved mixed forests (∼1800 stems/ha), respectively. Compared with the classical individual tree segmentation methods (marker-controlled watershed segmentation and point cloud region-growing algorithm), our method obtained comparable performance in low-density conifer stands and superior performance in the other stands. Furthermore, the region-merging algorithm could detect 10% more suppressed trees on average, which led to an apparent improvement in detection accuracy. The proposed algorithm provides a flexible segmentation framework that could be further improved by a different design that merges costs or applies multiscale segmentation with different stopping criteria.**

***Index Terms*— Hierarchical stepwise optimal, individual tree segmentation, light detection and ranging (LiDAR), region-merging, unmanned aerial vehicle (UAV).**

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I. Introduction

FOREST ecosystems are the main body of the terrestrial biosphere, which cover 3999 million hectares, approximately 30.6% of earth's land area [1]. Forests provide vital long-term ecosystem services to both humans and the surrounding natural environment, such as supplying food, water, and timber stock, contributing to biodiversity conservation and regulating the global climate system [1], [2]. To ensure that these functions are always well-preserved and maintained, both timely and accurate forest inventories are needed [3], [4].

Conventionally, forest inventories rely on field surveys that measure individual tree parameters [e.g., tree height, diameter at breast height (DBH), and crown width] within the established sample plots. This traditional method can obtain detailed information at the individual tree level but suffers from limitations due to high labor costs and timing and accessibility issues [3], [4]. Airborne light detection and ranging (LiDAR), which is an active remote sensing technology, is capable of capturing detailed and accurate three-dimensional (3-D) information of forest structures without spatial and temporal limitations; hence, LiDAR has become a powerful tool for characterizing and monitoring forest structures [5]–[7]. Depending on individual tree crown detection and delineation algorithms, detailed forest parameters at a single-tree level can be directly measured, which may function as a great complement to field measurements [8], [9].

Various algorithms have been developed for segmenting individual trees from the LiDAR data. Typically, most approaches convert 3-D point clouds into digital surface models (DSMs) or canopy height models (CHMs) and image processing techniques are applied in further steps to detect and delineate individual trees [9]. Generally, treetops are first identified using local maxima filtering [10], [11] before crown boundaries are delineated using various methods, such as the watershed algorithm [12]–[14], region-growing method [15]–[17], and valley following method [18], [19]. Because the final delineation results are highly dependent on the detected treetops, many previous studies tended to apply local maxima filtering with variable window sizes to solve the commission and omission errors caused by nonuniform crown sizes [10], [13], [20]. However, these approaches are often considered site-specific crown diameter-height allometry models, which are inappropriate for application in other sites or present undesirable performance at other sites due

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to the weak relationship between crown diameter and tree height [8], [21], [22].

Many methods focus on delineating crowns devoid of prior treetop detection to avoid falsely detected treetops, which influence the final crown segmentation results of a CHM. Liu *et al.* [21] developed a Fishing Net Dragging algorithm to refine the crown boundaries delineated by watershed segmentation and used a boundary classification model to merge the adjacent segments. Zhao *et al.* [23] considered the CHM a mountain-like topographic surface and used a hierarchical cross section among crowns in the vertical direction to segment individual crowns. Nevertheless, raster-based segmentation is significantly affected by image quality [24], [25]. A gridded CHM serves as a simplified representation of reality in the vertical direction and inherently loses details during the conversion from LiDAR point clouds to 2-D rasters [26], [27].

Considering the limitations of CHMs in accurately representing tree crowns, a growing number of studies have focused on directly segmenting individual trees by 3-D approaches, which can be classified as profile- and local density-based methods. The profile-based algorithms (distance-based clustering) tend to exploit the vertical profiles of crowns and use point clouds to segment trees in a top–down direction. Li *et al.* [28] developed a top–down region-growing algorithm that separates individual crowns sequentially based on the horizontal distance between tree crowns. Vega *et al.* [26] proposed multiscale dynamic segmentation processing to detect treetops and classify the points of the corresponding trees from the highest to the lowest point. Hamraz *et al.* [29] delineated individual trees by identifying local minima in different directions and iteratively executing the segmentation process from the tallest to the shortest point within a given area. Similar to the raster-based methods, these profile-based approaches also segmented trees based on defined treetops or valleys between crowns. However, overlapping crowns with a similar height merged their corresponding peaks, which led to a failure in the segmentation process [30].

Many studies have also applied local density-based algorithms (density-based clustering) that use local density features to detect and delineate trees. The first segmentation method that used the density distribution of point clouds was proposed in [31]. Rahman *et al*. [32]presented a density-of-high-points method for delineating crowns by assuming that the point density is high at the center of a tree crown and decreases toward the edge of the crown. Ayrey *et al.* [33] proposed a layer stacking method for segmenting trees, and it slices the entire point cloud into horizontal layers, isolates trees at each layer, and then detects individual trees on the stacked layer map. Furthermore, Ferraz *et al.* [34] regarded a crown apex as a local maximum in both density and height in point clouds. They applied a 3-D density-based clustering mean-shift algorithm to delineate trees. Recently, Ferraz *et al.* [35] and Hu *et al.* [36] improved the mean-shift clustering by adaptively calibrating the kernel bandwidth. However, these local density-based methods might be inefficient at addressing overlapping tree crowns, especially for high-density closed forests where clusters of dense points often appear within the conjunction areas [30].

Although many algorithms are effective for open canopy forests or forests with low stem density, where crowns have clear shapes with limited or no contact in between [37], [38], the success of tree detection is highly associated with forest conditions [9]. Under-segmentation may increase significantly in closed forests with high stem density, where crowns are often clumped together and overlapped and adjacent large trees oppress small trees [39]–[41]. An international benchmarking [41] and review in Zhen *et al.* [9] revealed that separating single crowns in high-density forests remains a challenge.

In addition, the under-segmentation of small trees is a common problem for delineating individual trees from the aerial LiDAR data [40], [42], [43]. Ferraz *et al.* [35] and Hamraz *et al.* [42] separated forest layers to detect individual trees at each layer to enhance the segmentation of understory vegetation. However, this method is inefficient at segmenting suppressed trees in high-density even-aged forests since the canopies could not be separated vertically. Other studies decomposed the segmentation process by exploiting hybrid approaches, which initially identified the constrained regions (including one or more trees) using a preliminary segmentation routine and then determined and independently segmented for each region containing more than one tree [40], [43]–[45]. However, the final results of such hierarchical approaches are dependent on the initial segmentation and detection under the initial segmented regions [42].

Other efforts have focused not only on determining the geometric characteristics but also on enhancing tree crown detection by combining the radiation characteristics of the full-waveform data [27], [46], [47] using the intensity information of leaf-off point clouds [48] or the spectral characteristics of the multispectral LiDAR data [49], [50]. Although additional optical information has been indicated to benefit individual tree detection, these methods are limited by data acquisition. The reflectance and calibration of the object and sensor system are not easily applicable to different cases [46], [48]. In contrast, the use of 3-D geometric characteristics of forest point clouds is potentially more robust for segmenting individual trees.

Recently, miniaturized laser scanning sensors on unmanned aerial vehicle (UAV) platforms have gained a positive reputation in forest inventory surveying since they have various benefits, such as increased point densities, decreased costs, flight flexibility, and simple operation [3], [51], [52]. An increasing number of studies have applied unmanned aerial vehicle light detection and ranging (UAV-LiDAR) to monitor and estimate forest structures, such as the tree height [53], [54], crown profile [55], leaf area index (LAI) [54], [56], volume [57]–[59], and aboveground biomass [57], [60]. Compared with airborne data, the high point density of the UAV-LiDAR data achieves reduced bias and standard deviations when measuring the crown structures and could significantly decrease the omission of trees [53], [61].

In this study, we proposed a novel 3-D approach for individual tree crown segmentation and applied and assessed the feasibility of the proposed algorithm using high-density UAV-LiDAR data under different forest densities. More specifically, the objectives of this study were to: 1) develop a

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region-merging framework for 3-D delineation of individual trees from LiDAR point clouds; 2) exploit a global merging cost of entire point clouds without specifying crown shapes and performing local searching; 3) explore the algorithm under different forest density and crown classes; and 4) compare its performance with that of the existing segmentation methods. This study will provide new insights to advance the 3-D segmentation of individual trees from LiDAR point clouds.

II. Hierarchical Region-Merging Algorithm

In this section, we present a novel algorithm for segmenting individual trees from 3-D point clouds, which was inspired by hierarchical region-merging methods in image segmentation [62]. Specifically, entire forest point clouds were first over-segmented into initial partitions, these partitions were then merged iteratively by a hierarchical stepwise optimal approach, and a given hierarchy was finally obtained. The proposed method performed a bottom–up hierarchical segmentation consisting of four main aspects: initial partition, hierarchical stepwise optimal region-merging, merging cost criterion, and stopping criterion, which are described sequentially in Sections II-A–II-D. LiDAR points were organized using a k-dimensional tree structure to accelerate distance searching in the algorithm [63].

A. Initial Partition

In 3-D point-based individual tree delineation, two inherent problems prevent the algorithm from segmenting efficiently: first, points are merely a result of discretization; and second, a large number of points require high computational costs. Inspired by the superpixels generated in image segmentation, which initially group pixels that perceptually belong together (with certain visual significance composed of adjacent pixels with similar texture, color, and brightness), and simultaneously heavily decrease the number of primitives for subsequent algorithms [64], we applied over-segmentation to point clouds to generate initial partitions and then used these partitions as inputs for region-merging. In this way, point-based segmentation could be adapted to a region-based approach for crown delineation. The feature derived from each region can be conducive to subsequent region-merging segmentation.

In this study, we applied mean-shift clustering to generate over-segmented initial partitions from point clouds. Mean-shift is a nonparametric clustering method for converging to local maxima of density in a certain feature space [65]. Airborne LiDAR point clouds for forest stands could be considered a multimodal distribution in which each given mode defined as a local maximum in density corresponds to a crown apex or a part of the crown [34], [47]. To make mean-shift vectors converge toward components of a crown to produce over-segmented initial partitions, we defined a spherical kernel with a bandwidth smaller than the crown radius. For a 3-D point set $\{X_i = (x_i, y_i, z_i) \mid i = 1, 2, \ldots, n\}$, the mean-shift vector $m_h(X)$ of a point X is defined as follows:

$$m_h(X) = \frac{\sum_{i=1}^{n} X_i g\left(\left\|\frac{X-X_i}{h}\right\|^2\right)}{\sum_{i=1}^{n} g\left(\left\|\frac{X-X_i}{h}\right\|^2\right)} - X \tag{1}$$

where g represents the kernel profile, and h is the bandwidth of the kernel. To attribute the highest weights to the points closest to the center [34], [36], the kernel profile g follows a Gaussian function:

$$g\left(\left\|\frac{X-X_i}{h}\right\|^2\right) = \begin{cases} \exp\left(-\frac{1}{2}\left\|\frac{X-X_i}{h}\right\|^2\right), & \text{if } \left\|\frac{X-X_i}{h}\right\|^2 < 1. \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

B. Hierarchical Stepwise Optimal Region-Merging

For given initial partitions, the hierarchical stepwise optimization (HSWO) approach proposed by Beaulieu and Goldberg [62] was applied for sequential region-merging in an iterative process. At each iteration, the merging cost measures $\text{cost}(Si, Sj)$ were calculated for all pairs of segments (Si, Sj), and the pair minimizing the measure $\text{cost}(Si, Sj)$ was merged for stepwise optimization.

In practice, we performed HSWO region-merging based on a region adjacency graph (RAG) $\text{RAG} = \{V, E\}$, where V is a set of vertices representing the clusters, and the neighboring relationships are represented by edges E [66]. The weight of the edge is denoted by the merging cost of the corresponding pair. The RAG-based region-merging is shown in Fig. 1. First, the initial RAG is constructed by initial partitions. In each merging step, the minimum weighted RAG edge is determined, the linked regions (vertices) are merged, and then the RAG is updated. This merging step is repeated until the stopping condition is satisfied. The merging cost and stopping condition are introduced in the following section.

Different from image segmentation, the adjacent relationships could be determined by shared pixel boundaries, and we defined the neighboring relationship by the minimum adjacent distance $d_{\min}$ of the two point sets $S_i = \{p_1, p_2, \ldots\}$ and $S_j = \{q_1, q_2, \ldots\}$ as shown in the following equation:

$$d_{\min}(S_i, S_j) = \min_{p \in S_i, q \in S_j} \{\text{dist}(p, q)\} \tag{3}$$

where $\text{dist}(p, q)$ represents the horizontal Euclidian distance between point p and point q. The edges are determined according to the following equation:

$$e_{i,j} = \begin{cases} 1, & d_{\min}(S_i, S_j) < d_{\text{thr}} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $e_{i,j}$ is the adjacency of S_i and S_j, d_{thr} is a distance threshold controlling the length of edges. $e_{i,j} = 1$ indicates that S_i and S_j are connected to each other; and $e_{i,j} = 0$ indicates that this edge fails. The details for the stepwise criterion and the stopping condition depending on the particular application are given in the following sections.

C. Merging Cost Criterion

In the HSWO merging procedure, the stepwise criterion $\text{cost}(Si, Sj)$ controls the merging order of all regions, which mainly affects the quality of individual tree segmentation through the region-merging method. Although the HSWO

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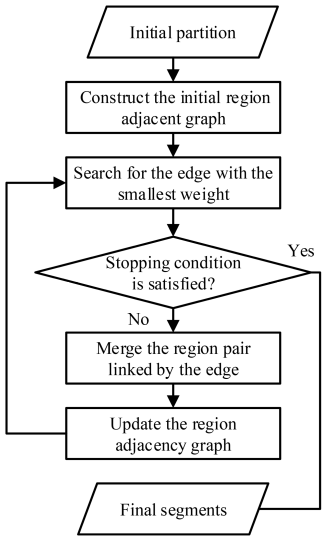


Fig. 1. Flowchart of the hierarchical region-merging based on the region adjacency graph.

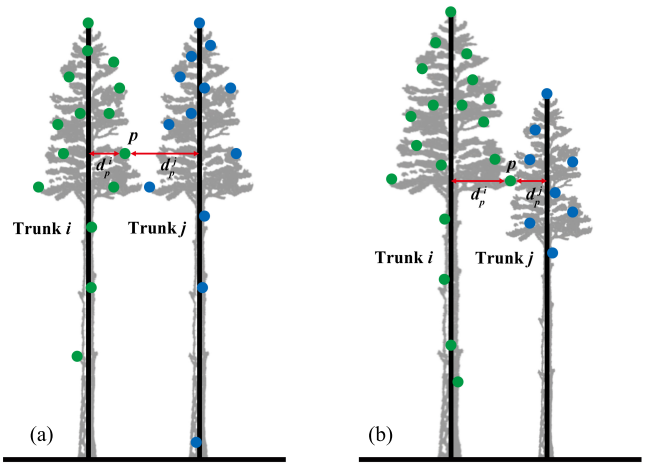


Fig. 2. Simplified illustration of the compactness-constrained global criterion. (a) Trees i and j are uniform in shape and size. (b) Trees i and j have different crown sizes. Green and blue dots belong to trees i and j, respectively. Trunks are presented by the solid vertical line from the top of the tree to the ground.

presents a local optimization procedure, it can be adapted to the global optimization case by an appropriate definition of a stepwise criterion [62], [67]. In this study, the compactness-constrained global model was developed to segment individual tree point clouds.

A schematic was presented to illustrate the compactness constraint criterion (Fig. 2). As shown in Fig. 2(a), assuming tree i and tree j have a uniform shape and size, the horizontal distance from each point on a single tree to its corresponding trunk is smaller than that to other trunks [see Fig. 2(a), $d_p^i < d_p^j$]. If we regarded the distance as an error term, then the compactness of cluster S_i could be presented by the sum of squared errors $E(S_i)$ as shown in the following equation:

$$E(S_i) = \sum_{p \in S_i} d_p^{i^2} \tag{5}$$

where d_p^i is the horizontal distance from point p to its corresponding trunk. A large $E(S_i)$ presents low compactness. Global shape compactness represented by the total sum of squared errors for all segments $S = \{S_1, S_2, \ldots, S_n\}$ could be used as a global criterion

$$G(S) = \sum_{S_i \in S} E(S_i). \tag{6}$$

The best segmentation for all trees pointed to the maximum global compactness, which minimized the total sum of squared errors $G(S)$.

The shape and size of individuals are highly varied in real-world forests. Thus, a direct comparison between horizontal distances is not appropriate as shown in Fig. 2(b), in which point p belongs to tree i but the distance from p to trunk i is larger than that to trunk j. The edge of a large crown will be falsely divided into adjacent smaller trees. To address this issue, we scaled the distance between points to the corresponding trunks by a weight function so that a large cluster had a high priority during region-merging. Herein, the number of points for a cluster was practically exploited as a scaling parameter. The global criterion $G(S)$ can then be calculated by the sum of the weighted squared errors

$$G(S) = \sum_{S_i \in S} \sum_{p \in S_i} w_{s_i} d_p^{i^2}, \quad \text{where } w_{s_i} = \frac{1}{N(S_i)^\alpha} \tag{7}$$

where w_{s_i} presents the weight for S_i; $N(S_i)$ denotes the number of points in S_i used for cluster size explanation; and α is defined as a scaling exponent to adjust the constraint of the number of points. Considering that trunks are not always visible in aerial laser scanning under leaf-on conditions, especially for dense forests [43], we used the xy coordinate of the highest point of a cluster to represent the position of trunks, which is commonly used to locate individual trees [9], [68].

In hierarchical region-merging, the goal of minimizing the global criterion $G(S)$ could be achieved by minimizing each term of the summation corresponding to the increase in G at each iteration [62]. An initial partition with n segments is defined as $S^0 = \{S_1^0, S_2^0, \ldots, S_n^0\}$. At the kth iteration, the algorithm merges two segments from the S^{k-1} partition to produce a new partition $S^k = \{S_1^k, S_2^k, \ldots, S_{n-k}^k\}$. As the number of segments is decreased by one at each iteration, S^k must contain $n - k$ segments. $G(S)$ at the kth $G(S^k)$ can be written as follows:

$$G(S^k) = G(S^0) + \sum_{t=1}^{k} \Delta G_t \tag{8}$$

where $\Delta G_t = G(S^t) - G(S^{t-1})$ refers to the increase in G at each merging. The minimization of the global criterion is represented by a sequence of stepwise optimizations that minimize each increase in ΔG_t in k iterations. Since the increment is determined by only the two merging segments in an iteration, the object of stepwise optimization is to search for the two segments S_i and S_j, whose merger $S_i \cup S_j$ produces the minimum increase in $G(S)$. The merging cost of S_i and

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S_j is defined in detail as follows:

$$\text{cost}(S_i, S_j) = \sum_{p\in(S_i\cup S_j)} \frac{1}{N(S_i \cup S_j)^{\alpha}} d_p^{ij^2} - \sum_{p\in S_i} \frac{1}{N(S_i)^{\alpha}} d_p^{i^2} - \sum_{p\in S_j} \frac{1}{N(S_j)^{\alpha}} d_p^{j^2} \quad (9)$$

where d_p^{ij}, d_p^i, and d_p^j represent the horizontal distance from a point p to its corresponding treetop in $S_i \cup S_j$, S_i, and S_j, respectively. $\text{cost}(S_i, S_j)$ is used as the weight on the edge between S_i and S_j in RAG-based region-merging.

D. Stopping Criterion

For an initial partition with N segments, the HSWO region-merging process iteratively executes $N-1$ times and produces $N-1$ different segmentation results. A vital consequence of this hierarchical structure for the stepwise optimization algorithm is to specify the level at which to stop the segment merging. Similar to the scale parameter in object-based image analysis, the stopping criterion controls the fineness of segmented regions, which could be defined by the merging cost or the number of segments in the partition [69]. To construct a robust stopping criterion under similar forest stands, the stopping criterion should monotonically increase within the merging process; thus, a stopping threshold only breaks the segment hierarchy by one time. In this study, the merging cost at the ith step was calculated as the stopping criterion τ_i for the optimization process

$$\tau_i = \max\left(\tau_{i-1}, \sqrt{\max(0, \text{cost}_i)}\right) \quad (10)$$

where cost_i refers to the ith merging cost in the HSWO region-merging process. A stopping threshold ST was defined to control the maximum merging cost. If τ_i exceeded the stopping threshold ST, then the region-merging stopped and the final segmentation was completed.

III. Experiments

A. Study Sites

Four study sites situated in two study areas, Mengjiagang Forest Farm (MJG) and Maoershan Forest Farm (MES), were selected for algorithm assessment (Fig. 3). MJG (130°32′42″–130°52′36″E, 46°20′16″–46°30′50″N) is located along the western fringes of the Wanda Mountain range in the northeastern part of Huanan County, Heilongjiang Province, China. The forest vegetation in the area is dominated by coniferous plantations, predominantly *Larix olgensis*, *Pinus koraiensis*, *Pinus sylvestris var. mongolica*, and *Picea asperata*. Three *Larix olgensis* sites (sites I–III) were selected according to different stand densities. Site I was a mature forest with low stand density (~700 stems/ha), a small amount of crown overlap, and a few suppressed trees. Site II was a near-mature forest with moderate crown overlaps and stand density (~1000 stems/ha). Site III was a middle-aged forest with high stand density (~2000 stems/ha) and large crown overlaps, causing a certain number of crowns to be suppressed by the surrounding trees under high competitive pressure. The terrain of the three sites is relatively flat and presents small elevation differences.

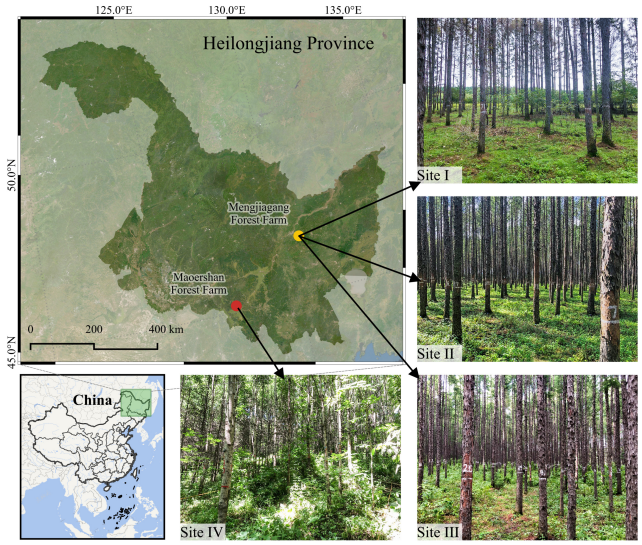


Fig. 3. Locations of the four study sites.

The fourth site (IV) is located in MES (127°18′0″–127°49′6″E and 45°2′20″–45°18′16″N), a temperate forest located along the western fringes of Zhangguangcai Mountain ranges in the northeastern part of Shangzhi City, Heilongjiang Province, China. This site was a coniferous and broad-leaved mixed forest dominated by *Larix olgensis* and *Fraxinus mandshurica*, with high stand density of approximately 1800 stems/ha. The crowns are often heavily sheltered and intertwined with each other, and broad-leaved trees usually have irregular shapes with many sub-crowns and shelter the surrounding conifers. The terrain in the site is relatively flat with slopes less than 10°. Examples of the four forest stands are shown in Fig. 3.

B. UAV-LiDAR Data Acquisition and Preprocessing

The UAV-LiDAR data were acquired in July 2019 and August 2018 for MJG and MES, respectively. The specifications of B. UAV-LiDAR surveys for these study areas are summarized in Table I. The raw UAV-LiDAR data were processed by coordinate transformation, strip clipping, and adjustment [56], [57]. The average point densities were approximately 200 pts/m^2 among sites I–III and approximately 180 pts/m^2 in site IV.

The georeferenced and merged point clouds were first clipped with a 5-m buffer around the boundary of each plot to capture complete crowns along plot boundaries [24], [70]. After the noise points were removed, the remaining points were classified into nonground and ground points using cloth simulation filtering [71], which was implemented in the open-source software CloudCompare (http://cloudcompare.org). Kriging interpolation was then applied to generate DTMs with a 0.5-m pixel size [72]. The terrain in the sample plots is relatively flat and presents small elevation differences. Then, the point clouds were normalized by subtracting the corresponding DTM value from the elevation of each point [28], [73], and

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TABLE I
UAV-LiDAR Acquisition Specifications for the Two Study Areas

Study area	Mengjiagang Forest Farm	Maoershan Forest Farm
LiDAR sensor	RIEGL mini VUX-1UAV	Velodyne Puck VLP-16
UAV platform	Feima D200	GreenValley GV1500
Flying altitude AGL (m)	80	60
Strip spacing (m)	80	25
Speed (m/s)	5.0	3.6
Pulse rate (kHz)	100	21.7
Maximum Range (m)	250	100
Maximum echo number	5	2
Scan angle (degrees)	±60	±15

points above 3 m were used for subsequent segmentation to eliminate the influence of low vegetation on the segmentation of individual trees.

C. Field Measurements

A total of 33 square plots (30 m × 30 m) were established in MJG in July 2019, including eight plots in site I, 13 plots in sites II, and 12 plots in site III. For site IV, three 50 m × 50 m plots were established in May 2018. All sample plots were evenly distributed within each site and avoided the edges of stands and large forest gaps. All trees with DBH values larger than 5 cm were measured in the sample plots. A total of 5510 live trees were observed among the four sites, and the field measurement attributes for each site are summarized in Table II. In the field survey, the crown width in the four cardinal directions and DBH were measured using metallic tapes and diameter tapes, respectively, while the height of the first live branch position (at the crown base) and the total tree height were measured using a Vertex IV instrument (Haglöfs, Sweden). The coordinates of the trees were also recorded by measuring their relative position to the corner of plots. In addition, the geographic coordinates of the individual trees and four corners of each plot were thoroughly measured with a real-time kinetic (RTK) global navigation satellite system (GNSS) (UniStrong G10A, China). A number of trees in areas with a poor GNSS signal were georeferenced by their relative coordinates.

D. Accuracy Assessment

To quantitatively assess the accuracy of the segmentation results, we matched the detected trees located within the plot boundaries with the *in situ* reference measures based on spatial locations and height differences [27], [74], [75]. The position and height of a detected tree were determined by the position and height of the highest point in a segment [27], [75]. The segmented trees were assigned as candidates of a reference tree if the horizontal distance (ΔP) from the reference was less than the corresponding crown radius (with an upper limitation of 3 m) and the height difference (ΔH, absolute value) from the reference was less than 20% of the top height of the plot [75]. A unique candidate or the closest one among multiple candidates was defined as a match with the reference tree [24], [76]. Segmented trees without a link to the references were considered false positives (commission errors). Conversely, reference trees that were not matched to any segmented trees were classified as false negatives (omission errors).

The accuracy of measurements in terms of precision (Pr), recall (Re,) and F-score (F) was calculated to further evaluate the segmentation performance [28]

$$\mathrm{Pr} = \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}} + N_{\mathrm{FP}}} \tag{11}$$

$$\mathrm{Re} = \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}} + N_{\mathrm{FN}}} \tag{12}$$

$$F = 2 \times \frac{\mathrm{Pr} \times \mathrm{Re}}{\mathrm{Pr} + \mathrm{Re}} \tag{13}$$

where N_{TP}, N_{FN}, and N_{FP} represent the number of matching pairs, omissions, and commissions, respectively; Pr and Re represent the correctness of detected trees and the detection rate of reference trees, respectively; F is the harmonic mean of Pr and Re, which describes the overall accuracy of the segmentation algorithm.

In addition, the positioning, height, and crown radius absolute deviation (ΔP, ΔH, and ΔR) from the segments to references of the matched cases were further used as an evaluation of extracted single-tree parameters [74], [77].

Previous studies have suggested that canopy structures have a considerable influence on the performance of segmentation methods [39], [41]. Hence, we referred to [41], who classified reference trees into crown classes to perform a stratified assessment under different canopy structures. A classic tree classification method [78] divides individual trees into five classes: dominant, codominant, intermediate, suppressed, and dead trees. All the measured living trees were classified into four classes according to the characteristics of plantations in our study area. The qualitative definitions are described as follows, and the schematic is presented Fig. 4.

1) *Dominant*: The highest tree in the neighborhood (5-m buffer around the target tree in this study) as shown in Fig. 4(A_2)] or an isolated tree with no neighbors within the 5-m buffer [Fig. 4(A_1)].
2) *Codominant*: The nonhighest tree in the neighborhood with the largest overlap part less than 50% of the target crown radius [Fig. 4(b)].
3) *Intermediate:* The nonhighest tree in the neighborhood with the largest overlap part exceeding 50% but less than 100% of the target crown radius [Fig. 4(c)].
4) *Suppressed:* The tree with growth suppressed by the surrounding trees with the largest overlap exceeding the target crown radius [Fig. 4(d)].

All assessment metrics were further applied to evaluate the accuracy of the segmentation method in each crown class at the stand level. The commission errors are linked without any corresponding reference trees, and the crown classes of over-segmentations are assigned to the nearest references [41].

E. Comparative Analysis

To demonstrate the effectiveness of the proposed region-merging method, we compared our algorithm with two leading

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TABLE II
STATISTICS OF STAND VARIABLES FOR FOUR STANDS

Stand	Number of plots	Stand density (stems/ha)			DBH (cm)			Tree height (m)			Crown radius (m)		
		Range	Mean	Sd	Range	Mean	Sd	Range	Mean	Sd	Range	Mean	Sd
I	8	489-855	653	118	7.2-34.8	22.0	4.7	13.3-28.9	22.0	2.4	0.3-3.7	1.7	0.5
II	13	933-1367	1058	114	10.2-35.1	18.4	4.1	14.0-28.0	22.2	2.3	0.2-3.0	1.4	0.4
III	12	1722-2577	2091	270	5.1-25.0	12.5	3.2	6.1-21.3	15.6	2.1	0.3-2.6	1.2	0.3
IV	3	1612-2224	1849	328	5.0-25.4	11.7	2.9	5.7-20.0	13.9	1.7	0.4-2.8	1.2	0.4

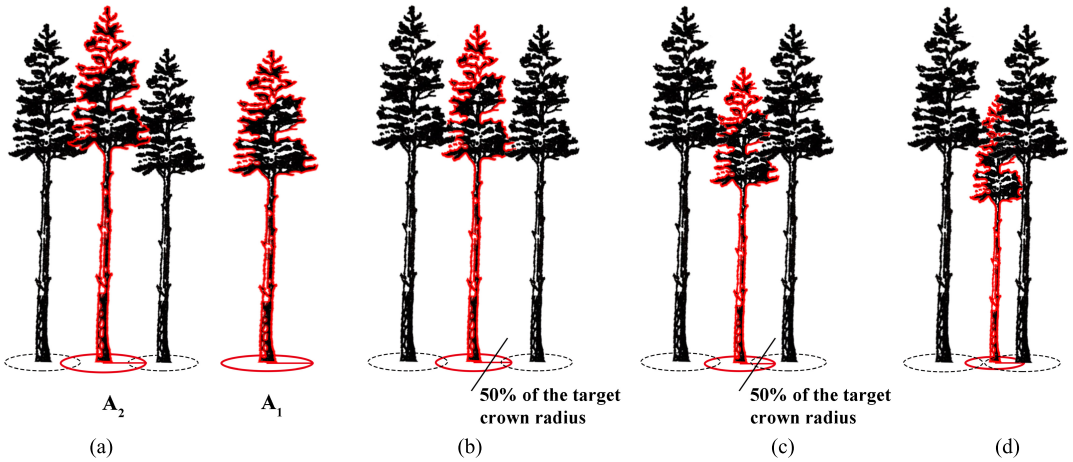


Fig. 4. Illustrations of crown classification. Trees in red boundary refer to the target trees, and solid red circles are shown as crown radii. Black trees refer to the neighbors of the target trees, and dashed black circles are the crown radii of the neighbors. Solid black lines represent the split lines of 50% of the target crown radius. (a) Dominant. (b) Codominant. (c) Intermediate. (d) Suppressed.

methods: the marker-control watershed segmentation (MCWS) and the region-growing method developed by Li *et al.* [28], which were commonly applied in practical application and comparative studies [41], [79]–[82].

MCWS is a CHM-based segmentation method whose performance is strongly affected by data pits in the CHM [24], [76]. The pit-free method proposed by Hao *et al.* [24] was therefore applied for CHM generation. Local maxima filtering was then applied to detect tree apexes, and MCWS was performed to delineate crown boundaries using the tree apexes as markers [13], [53]. Since the applied window radius of local maxima filtering affects the detection results, we adjusted this parameter to a range from 0.3 to 1.5 m for each stand and finally set it to 1.2, 1.0, 0.8, and 0.8 m for sites I–IV, respectively.

The point cloud-based region-growing method exploits the horizontal spacing between the target point and classified points to identify and delineate trees individually and sequentially from the point cloud [28]. In this approach, a threshold distance needs to be set that is adaptable to the structure of the forest [28], [83]; thus, it was also optimized in the range [0.3, 1.5] for each stand and set to 1.3, 1.0, 0.8, and 0.7 m for sites I–IV, respectively.

Both the methods were applied to the same dataset as the proposed method. The assessment metrics Pr, Re, and F were applied to evaluate and compare the individual tree detection accuracy. To further explore the differences in the number of correctly detected trees (TP) between region-merging and the comparison methods, we conducted chi-square tests for the TP relative to the number of references and TP relative to the number of detected trees (Re and Pr) in each stand.

IV. RESULTS

A. Sensitivity Analysis

In this study, a total of 5510 field-measured trees within 36 plots of four sites were used to verify the efficacy of the region-merging method. To quantify the sensitivity and tune the optimal parameters at the stand level, we first performed a sensitivity analysis by varying one control parameter at a time and holding the others at optimal values. For the initial partition, bandwidth h was set from 0.1 to 1.5 m, with an increment of 0.1 m. The neighbor distance threshold d_{thr} was varied from 0.1 to 1.5 m, with an increment of 0.2 m. The scaling parameter for adjusting the merging cost was varied from 0 to 1.0, with an increment of 0.1. ST was set within the ranges [1.0, 6.0], [2.0, 7.0], [3.0, 8.0], and [1.0, 6.0] using the same step of 0.1 for sites I–IV, respectively. The sensitivity analysis results of each parameter assessed using the mean F-score for each stand are illustrated in Fig. 5. The region-merging method was mainly sensitive to the stopping threshold ST and presented clear-cut distinctions for the different forest stands, while the other parameters had minor influences (F-score $< 1\%$) as long as their values were set to range from $h \in [0.3, 0.6]$, $d_{\text{thr}} \geq 0.5$, and $\alpha \in [0.5, 0.7]$ in either stand. Consequently, ST was set to 5.3, 3.6, 2.5, and 2.9 for sites I–IV, respectively. For the others, h, d_{thr}, and α were set as fixed values of 0.5, 0.5, and 0.6 for all stands because they represented the best compromise and, more importantly, were robust and universally applicable to all the reference data. The detailed impact of the parameters on the segmentation performance is discussed in Section V-B.

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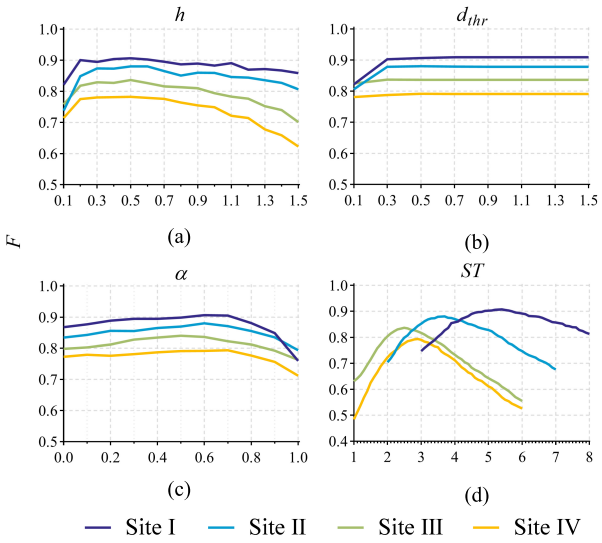


Fig. 5. Sensitivity analysis for parameters in the hierarchical region-merging algorithm. (a) Bandwidth h, (b) neighbor distance threshold d_{thr}, (c) scaling parameter α, and (d) stopping threshold ST. F represents the mean F-score for each stand.

B. Evaluation and Comparison

Examples of the segmentation results for different stands are presented in Fig. 6. It is visually apparent that most of the segmented trees are relatively complete and compact. Fig. 7 shows a qualitative comparison of the tree crown segmentation obtained for a high-density coniferous stand (site III) by different methods. The region-merging method could visually achieve fewer omissions (FN) than the MCWS and region-growing algorithm. The subgraph of the 3-D point clouds shows that the segmentation method was able to delineate the crowns without obvious treetops or valleys between crowns.

All the comparison results were summarized at the stand level and calculated separately for each crown class, as shown in Table S1. Fig. 8 shows the results of crown delineation under the four forest stands and chi-square test results. In general, for the coniferous sites (I−III), Pr, Re, and F of the three methods decreased with the increasing density of stands, and Pr was always higher than Re for both the methods, meaning that all the methods tended to produce more omissions than commissions in coniferous sites, while three methods performed differently in the mixed forest (site IV). By comparison, the proposed algorithm was more stable than the others across different stands, obtained relatively balanced Pr and Re values, and achieved the highest overall accuracy (F-score) among the three methods. The proposed region-merging method was able to detect 78%–86% of the reference trees across all stands, while the tree detection precision (Pr) ranged from 81% to 96%. The overall accuracy (F) was 91%, 88%, 84%, and 80% for sites I-IV, respectively.

In the low-density coniferous stand (site I), all the methods performed comparably and satisfactorily ($F > 90\%$), with the region-merging method exhibiting a small improvement over the other methods. Our method mainly showed a significant improvement in controlling omission errors in the medium- and high-density stands (sites II and III), which increased by 11% and 13% compared with the MCWS method and 7% and 9% compared with the region-growing method, respectively. The region-merging method also presented competitive Pr in sites II and III. Although our method had only a slightly lower Pr than the others (1%) for the high-density stand (site III), the results of the chi-square test showed no significant difference in the constituent rates of TP in the number of detected trees. The region-merging method could perform better than the other methods in coniferous sites, especially in the high-density stands. For the mixed-species site (IV), under- and over-segmentation become the main error sources of the MCWS and region-growing methods, respectively. The region-merging method exhibited more stable and higher Re and Pr than other methods and thus performed the best in the mixed forests.

Specifically, the radar plot in Fig. 9 comprehensively compares the accuracy of the different methods for all crown classes and forest stands. The center of the radar plot represents $\text{Pr}/\text{Re}/F = 0$, while the outer boundary represents the highest accuracies. In general, the proposed method almost enveloped the others in radar plots, indicating a better performance in stratified evaluation. Specifically, our method increased the average Re and F by approximately 10% and 11%, respectively, for suppressed trees compared with other methods.

With respect to a stand with different crown classes, the three methods exhibited no significant difference in Pr, Re, and F in the dominant, codominant, and intermediate trees in site I, but the region-merging method achieved the highest Re and F in suppressed trees, which was 15% and 16% higher than MCWS and 11% and 15 than region-merging methods, respectively. Our method yielded a slightly worse Pr than MCWS but better Pr than the region-growing method for suppressed trees. Considering the relatively small number of detected suppressed trees in the low-density stand, there was little difference in the number of commission errors (≤ 3 in average). In sites II and III, our method presented significant improvements by an average of 14% and 10% Re compared with the MCWS and region-merging methods, respectively, in all crown classes except for the dominant trees. With regard to Pr, three methods performed comparatively well across different crown classes. Our method achieved comparable F for dominant trees and exhibited superior performances for other crown classes, especially for the suppressed trees, for which the value increased on average by 14% and 10% compared with the MCWS and region-growing methods, respectively. For the mixed forests, the region-growing method generally achieved the lowest Pr, especially for the dominant trees (13% lower than with region-merging), while MCWS achieved the lowest Re, especially for the suppressed trees (8% lower than with region-merging). By comparison, the region-merging method always achieved the best F in each crown class.

In the same crown class across different stands, the most obvious aspect is the depression in the Re of suppressed trees for all sites, implying that the suppressed trees could not obtain competitive detection rates with other classes in every stand and method. In relatively complex stands with high densities (sites III and IV), the Re of suppressed trees

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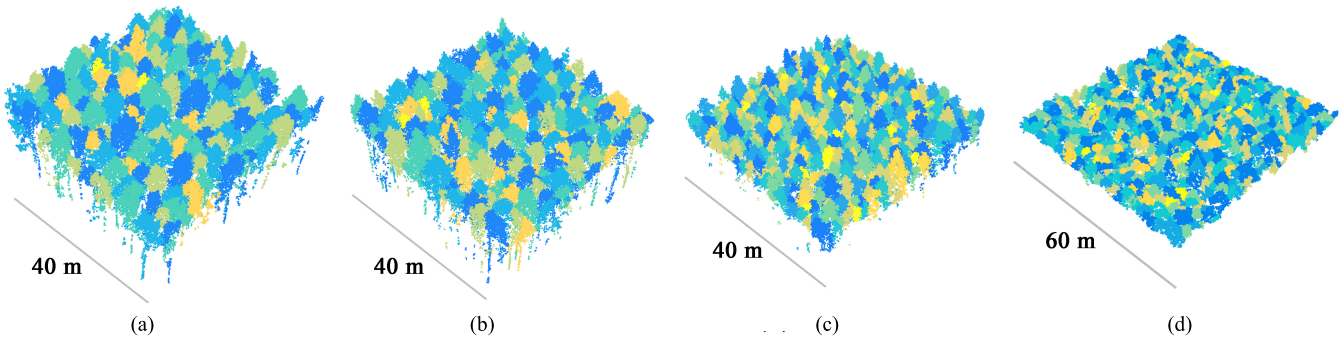


Fig. 6. Examples of the segmentation results for different sites. Adjacent trees were rendered with different colors. (a) Site I. (b) Site II. (c) Site III. (d) Site IV.

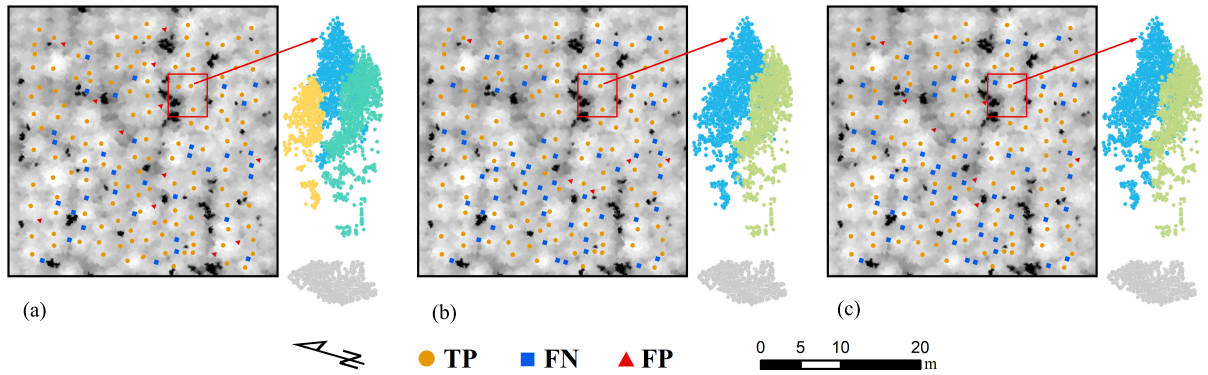


Fig. 7. Qualitative comparison of the region-merging, marker-controlled watershed segmentation (MCWS), and region-growing methods. TP represents the position of a correctly segmented tree which was matched with a reference tree; FN represents the position of a reference which was omitted by the segmentation algorithm; and FP represents the position of a segment which did not match with a reference. (a) Region-merging. (b) MCWS. (c) Region-growing.

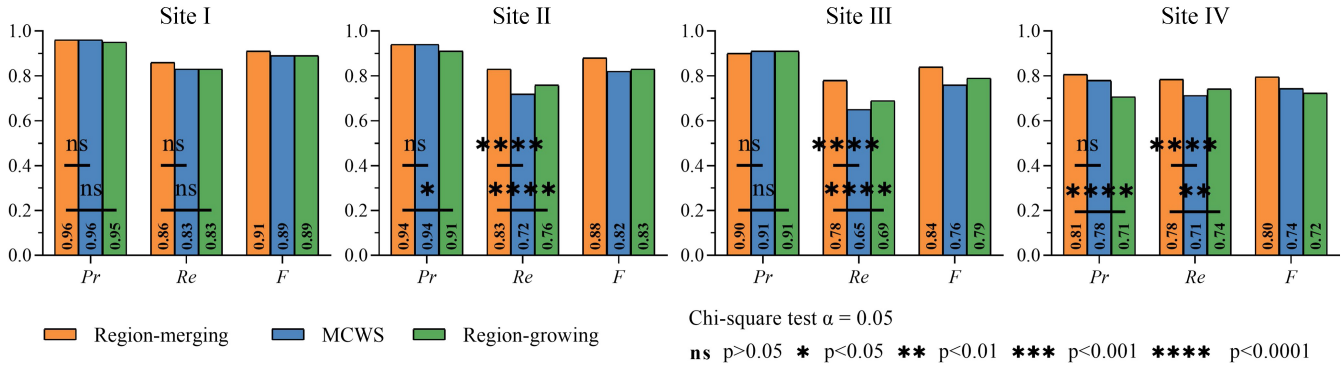


Fig. 8. Comparison results for the proposed region-merging algorithm and comparison methods: marker-controlled watershed segmentation (MCWS) and region-growing methods in all four stands.

was higher than that in the other stands. Among coniferous stands (sites I–III), the F decreased the most for the dominant trees in the high-density stand compared with the other stands, while the codominant and intermediate trees exhibited a smaller decrease. These findings are mainly associated with the decreased Pr with the increased stand density and the lowest Pr in the dominant trees of this stand. Furthermore, Pr of dominant trees in the mixed forest (site IV) showed a distinctly lower accuracy than in the coniferous sites.

The positioning, height, and crown radius differences (ΔP, ΔH, and ΔR) between the references and estimates from different segmented results were plotted across crown classes in each stand (Fig. 10 and Table S1). In general, our method achieved mean absolute error of positioning of 0.33, 0.32, 0.35, and 0.41 m; height of 1.05, 1.11, 0.85, and 0.64 m; and crown radius of 0.71, 0.44, 0.35, and 0.36 m for sites I–IV, respectively. It can be seen that all the three methods achieved comparable positioning and height estimation accuracy in the total columns, as shown in Fig. 10, in which the mean differences in ΔP and ΔH between methods were all less than 0.1 m and showed a similar distribution; the point cloud-based methods showed a slightly better performance than the raster-based MCWS for crown radius estimation. Similar results are shown for every crown class except for the suppressed class.

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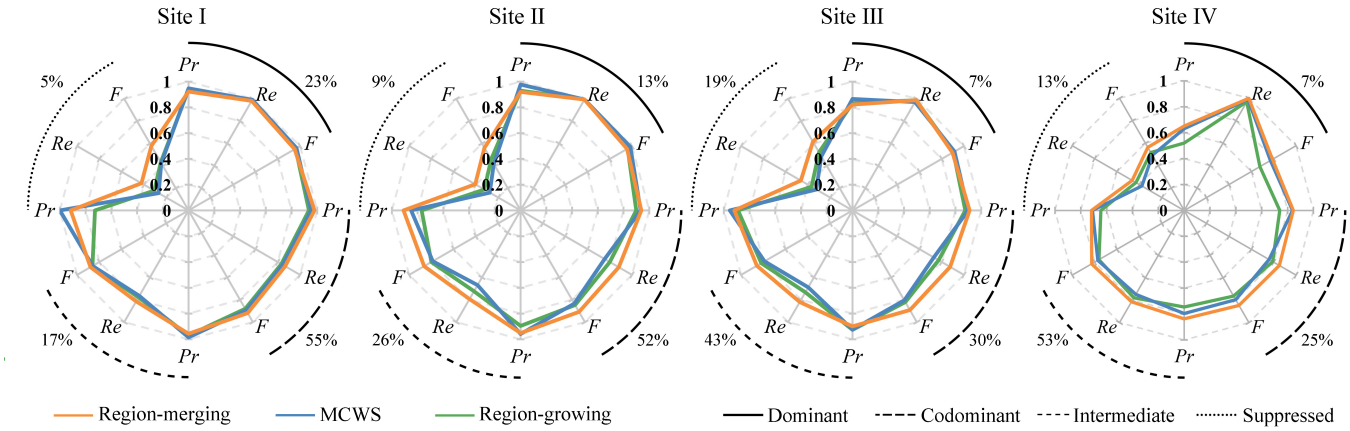


Fig. 9. Radar plot showing the precision (Pr), recall (Re), and F-score (F) for different classes in all stands using region-merging, MCWS, and region-growing algorithms, respectively. The proportions of each crown class are labeled beside each graph.

Furthermore, our method achieved slightly worse accuracy in the positioning and height assessment in the suppressed trees, but it is worth noting that the number of matched samples used to evaluate the parameter estimation differed in each method. All the above-mentioned quantitative comparison results were summarized and listed at the stand level and calculated separately for each crown class; see Supplemental materials for details.

V. Discussion

A. Performance of Individual Tree Segmentation Algorithm

Our study proposed a novel hierarchical region-merging method for delineating individual crowns from 3-D LiDAR point clouds, and it was mainly motivated by the reduced effectiveness of the traditional methods of detecting the characteristics of peak-like uplift. The proposed algorithm was tested in various forest conditions (coniferous forests of different densities and a coniferous and broad-leaved mixed forest) and compared with two leading individual crown delineation methods (MCWS and region-growing algorithms). To our knowledge, most studies on individual crown delineation have focused on forests with densities that range from 100 to 600 stems/ha [28], [37], [38], [50], while crowns are often overlapped and clumped together, and small trees are suppressed by the surrounding trees under high competitive pressure in high-density forests, making it difficult to isolate individual trees [9]. In this study, the results showed that the proposed method significantly improved the segmentation performance for high-density coniferous and coniferous and broad-leaved mixed forest compared with the traditional methods; thus, the proposed method may provide opportunity for segmenting individual trees with high stand density.

Generally, the segmentation accuracy of coniferous forest decreased with the increase in stand density, which is consistent with [68], while the accuracy in coniferous and broad-leaved mixed forest with similar density was not as good as in coniferous forests, as shown in [81]. According to the stratified assessment, this decreased accuracy as the forest density increased could be attributed to the relatively low detection rates of suppressed trees, and the proportion of suppressed trees increases with the stand density in coniferous forests (Fig. 9). In addition, under-segmentation may represent a main error source for our method in coniferous stands, as in many previous studies [20], [23], and it could also be explained by the large number of suppressed trees that were recorded as a cluster of surrounding trees. Nevertheless, the detection rate of our method was 10% for suppressed trees, outperforming both the MCWS and region-growing methods, thus improving the Re and F in the high-density forests. Because our method takes advantage of the global compactness of the entire forest landscape instead of the local detection of peak-like uplift crown features, it is beneficial for segmenting suppressed trees that do not present obvious treetops or interactions with other trees, which is shown via a visual inspection (Fig. 9). Moreover, the commission errors in the mixed forests were obviously higher than those in coniferous forests, especially for dominant trees, and resulted in lower accuracy than in coniferous stands. Previous studies have discussed the challenge of accurately delineating broad-leaved trees [29] because the branches of crowns or sub-crowns of these species were more likely to be mistaken as individual trees, as broad-leaved trees have larger and more irregular crowns [23]. By comparison with other methods, our method improved the Pr in every crown class because our method did not detect the valleys between crowns and thus was insensitive to the sub-crowns of broad-leaved trees.

In Section IV-B, Re of the suppressed trees in the relatively high-density stands (both coniferous and coniferous and broad-leaved stands) was shown to be higher than that of the other types of stands, while the detection of dominant trees was significantly lower in such forests in our method. These findings are still logical because parameter selection is aimed at optimizing the F-score, which is the harmonic mean between Re and Pr. When a large proportion of trees are suppressed in high-density stands, optimization of the F-score tends to compensate between commission and omission errors, which increases the Re at the expense of Pr. Larger crowns in

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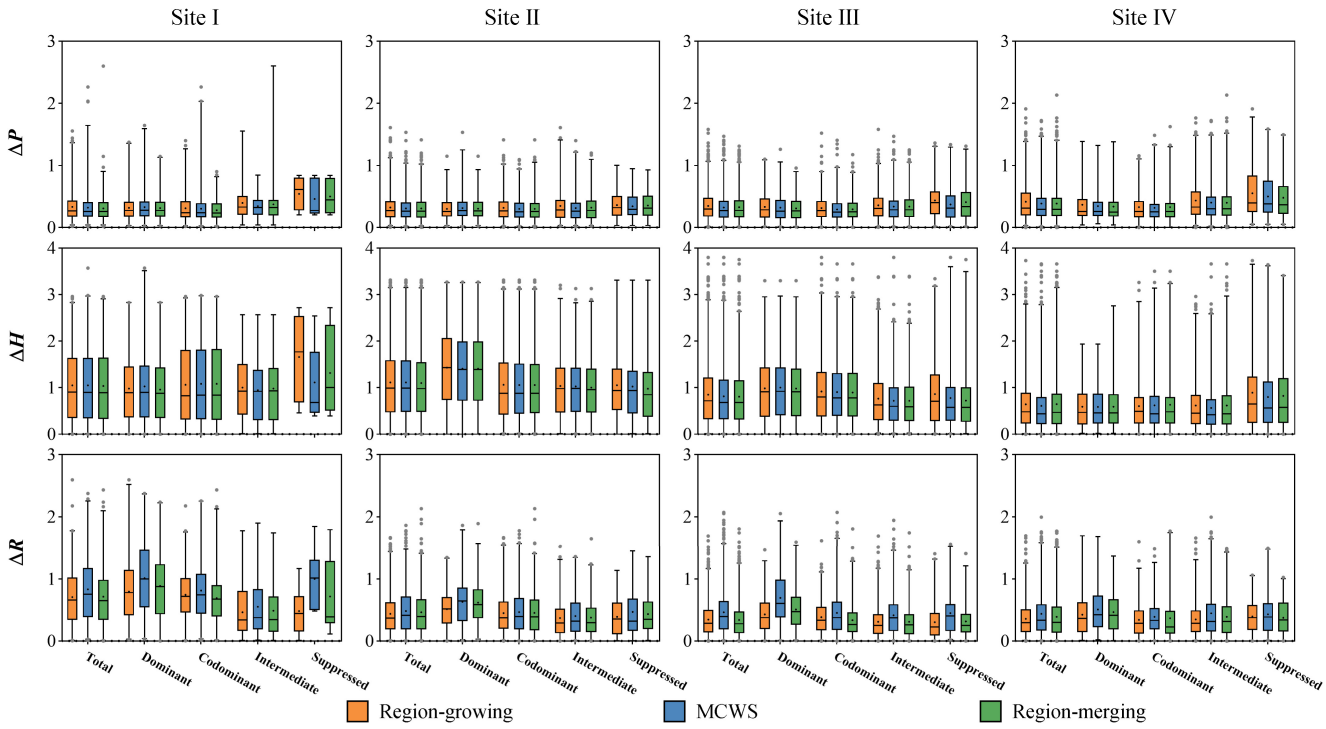


Fig. 10. Comparison of positioning and height differences (ΔP, ΔH, and ΔR) between the reference and estimates from the region-merging, MCWS, and region-growing algorithms across the different crown classes.

dominant positions are more likely to be divided into multiple crowns to increase the global compactness, which aggravated the over-segmentation of broad-leaved trees, whereas suppressed trees have more potential to be correctly segmented under this compensation.

Regarding the validation of the detected tree parameters, the different methods presented limited differences in the tree attribute estimations, which is consistent with [74] and [68]. Only the matched cases that were previously selected under the constraint of position and height difference were assessed. Furthermore, both the position and height were determined by the treetops from segmented point clouds, which were extracted from nearly the same locations for the correctly matched segments among different methods, as in [68]. Typically, delineated individual crowns will underestimate the crown radius/diameter because the top–down scanning has a poor ability to detect the intersecting area [80]. By comparison, the point-cloud-based methods (region-merging and region-growing) achieved a better but limited performance in crown radius estimation than the raster-based method, which was corroborated by Hadas *et al.* [37]. This result indicated that point-cloud-based segmentation has the potential to provide a more detailed and accurate estimate of crown width, but the improvement is not significant as the CHM we used for segmentation has a high resolution of 0.1 m. In fact, the performance between methods presented more pronounced differences for tree detection than for tree attribute estimation [68], [74]. Our method presented a slightly worse attribute accuracy for the suppressed trees than the comparison methods, which was likely because of the greater number of matched suppressed trees. As mentioned above and in Fig. 7, our method could segment individuals without obvious treetops, which may lead to larger deviations compared with field measurements due to the lack of treetops from segmented point clouds. Although the segmentation of suppressed trees may be incomplete and lead to deviations in parameter estimation of individual trees, it may decrease the bias in the aggregation of forest parameters from individual trees to the plot or stand level, such as stand biomass and carbon storage estimations from LiDAR data [15], [84], which may not only increase the detection rate of suppressed trees but also contribute to LiDAR applications.

Besides quantitatively comparing with two classic methods, we, herein, also take a qualitative comparison with other existing methods [47], [85] to highlight the novelty and applicability. Similar to their studies [47], [85], our method starts with initial point clusters instead of each point, allowing the creation of visually meaningful entities to reflect geometric characteristics and reduces the number of primitives for subsequent segmentation compared with segment individual trees at the point level. But there are three main differences between the approaches proposed in our and their studies. First, compared with previous studies that applied the mean feature of clusters as input for the following segmentation, our methods fully used the geometric feature of each point in each cluster [i.e., (3), (5), and (9)]. Second, unlike their studies which applied a recursive binary normalized cut (Ncut) algorithm for subsequent segmentation which is a top–bottom clustering approach [47], [85], the proposed HSWO region-merging method is a bottom–up approach for generating final

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segments. These two approaches are fundamentally different from the perspective of clustering. Furthermore, Ncut starts with a matrix of similarity measures between all possible pairs and uses the eigenvectors of the matrix to recursively subdivide it [27], [47], [85], while our region-merging approach dynamically calculated global merge cost in an iteration and merged the pairs of the minimum of the global cost increase. As previously reported by Lee *et al*. [79] and Zhong *et al*. [86], Ncut applied the second smallest eigenvector for subdivision which often performed well for a cluster containing two distinct parts, but may discard information from subsequent eigenvectors and forcibly cut point clouds into two parts in multitree point set case, and thus may damage the local completeness of a single tree. In addition, when applying Ncut, treetops or trunk detection were often first applied to provide prior knowledge for similarity matrix calculation [27], [47], [79], [85]. The final segmentation would inherently be influenced by the tree position detection. By comparison, the proposed method treats each segment as potential individual trees and applied an HSWO to avoid the local compactness reduction caused by a global optimism, and thus benefited more for merging over segments. Furthermore, the proposed method does not require any prior knowledge of the crown shape and size, which takes advantage of the global compactness of the entire forest scene to avoid local searching. More importantly, the proposed bottom–up segmentation framework could be further improved by a different design that merges costs or applies multiscale segmentation with different stopping criteria.

B. Parameter Analysis

The proposed segmentation scheme relies on appropriate parameters. Testing of different parameter combinations based on practical experience and the proposed sensitivity analysis showed that the ST mainly influenced the performance of the region-merging method. ST directly controlled the number and size of segments, which was similar to the scale parameter in traditional multiscale image segmentation [87]. In Section IV-A, the optimal ST was shown to increase with stand density. A high ST would lead to more segments being merged, thus indicating large crown clusters and few subdivisions; therefore, it was considered to be most suitable for stands with low density. Because of the difficulty performing quantification based on prior information, we recommended recording the entire segment hierarchy, including all possible stopping thresholds for each plot. Users can dynamically assess the performance and select proper segmentation results from the merging process rather than re-execute the entire program (supplementary video).

For other parameters, h controlled the initial over-segmented results and affected the final segmentation of individual trees. The under-segmentation produced in initial partitions could not be eliminated in the subsequent region-merging process, while the over-segmentation results could be gradually merged. The detection rates (Re) for the initial partitions, as shown in Fig. 11, were applied for auxiliary interpretation. From Fig. 11, an oversized bandwidth significantly reduced the Re as expected, which means that high omission error

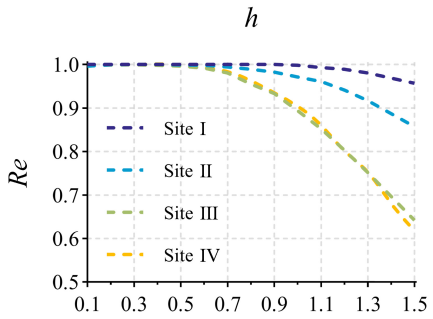


Fig. 11. Recall (Re) for initial partitions using different bandwidths h in mean-shift clustering.

rates occurred. In fact, small trees may be (omitted) merged into adjacent trees when the bandwidth is larger than its crown radius [34]. Therefore, the final segmentation accuracy decreased with increasing under-segmentation in the initial partitions. The sensitivity analysis in Section IV (Fig. 5) also showed that a relatively small h (e.g., $h = 0.1$) could not achieve desirable accuracies since a small h produces a small cluster, which makes it difficult to explain the local characteristics. Meanwhile, considering that the evaluation of too many regions in the initial partitions caused a large computation burden, we recommend selecting a larger bandwidth on the premise of few omission errors in the initial partitions.

In the region-merging process, d_{thr} controls the number of edges of vertices on the RAG, which should be defined as the maximum horizontal distance between two points within a single tree. A small value would reduce the adjacent regions and cause over-segmentations, while continuously increasing d_{thr} will dilate the neighbors of vertices but will not affect the region-merging process. In addition, α is a scaling constant that controls the merging cost and merging priority. Fig. 12 reflects the specific impact of α. After fixing the number of segments, the scaling constant of 1.0 causes the large cluster to gain the points belonging to the neighboring trees, as shown in Fig. 12(c). The value of $\alpha = 0.6$ indicated a relatively desirable performance [Fig. 12(b)]. In contrast, a smaller scaling constant, e.g., $\alpha = 0$, results in the edges of the large trees being split into adjacent small trees to obtain more uniform crown sizes [Fig. 12(a)]. This result compensated for the differences in the variety of crown sizes. A large alpha will potentially become preferable for stands with large crown differences. Many previous studies applied field-based or region-wide allometry relationships between tree height and crown width to estimate the size of the crown [8], [13], [20], [88]. However, our data were relatively concentrated, and the variation in the crown size was difficult to explain using only the tree height from the field measurements. The scaling function for the distance measurement in the merging cost calculation may be better approached with an algometric relationship than the power law of the number of points without a biological basis in future research. It could also be meaningful to estimate the crown size of a given cluster for weighting the distance measurement $E(S_i)$ (see Section II-B) based on metabolic scaling theory, such as in [89] and [82].

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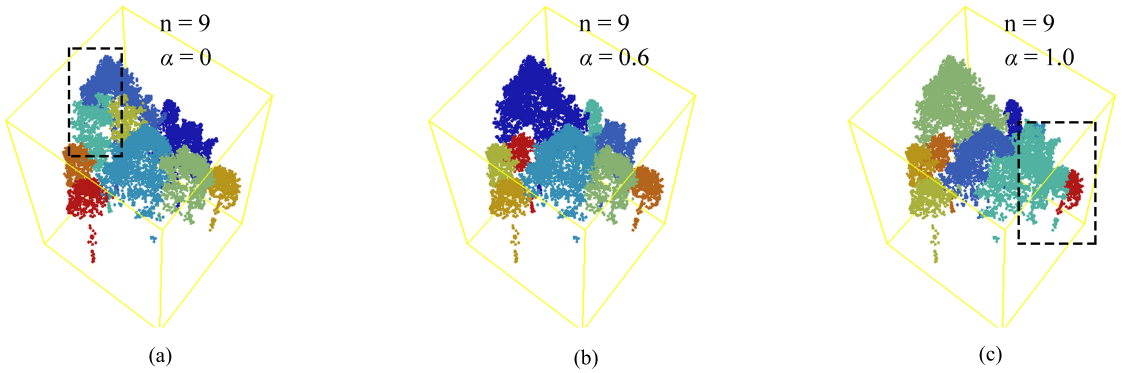


Fig. 12. Examples of segmentation results with a constrained number of segments ($n = 9$) and different α values. (a) $\alpha = 0$. (b) $\alpha = 0.6$. (c) $\alpha = 1.0$.

C. Potential Limitations and Outlook

Although the current experiments tested the proposed algorithm under various forests conditions, some potential influences and limitations should be stated for further investigation. As with most point-cloud-based algorithms [28], [34]–[36], [50], our method applied normalized point cloud as input because it is easy to eliminate the interference of ground points and understory vegetation on individual tree segmentation, and further, this approach can directly obtain the absolute height trees. However, the topographic correction may change the actual shape of the crowns and affect the performance, as discussed in [26], [90], and [91]; this correction should be considered for application in some steep areas. In addition, our study applied the power function of the number of points in regions as the scaling function for the merging cost. The partial excessive overlaps may decrease the weight in a cluster for merging cost measurement and thus affect the merging priority, potentially causing errors in final segmentation, especially for low-density point clouds and large α. Nevertheless, the application of UAV-LiDAR to forest analysis is often intended to obtain relatively uniform overlaps of point clouds for subsequent metrics generation and the partial excessive overlaps are often be clipped by post-processing.

Similar to previous individual tree segmentation algorithms, the main limitation of the proposed algorithm is that it is not a parameter-free method. The performance was sensitive to the parameter ST, which should be tuned in new applications with other forest types and different data acquisition methods. Nevertheless, the stopping condition could allow the algorithm to obtain segmentation results for any number and size. Furthermore, it presents a new insight for point cloud segmentation, and the bottom–up hierarchical segmentation could record the whole merging process to conveniently select the proper segmentation results (as shown in the supplementary video).

Considering computational performance, herein, all algorithms were run on a desktop computer with a 3.2-GHz Intel Core i7-8700 CPU and 16-GB RAM. Taking a 50-m × 50-m plot in site IV with a point density of approximately 180 pt/m^2 for a time consumption comparison, it took approximately 350 and 300 s for the point-cloud-based region-merging and region-growing methods, respectively, while the raster-based MCWS only took approximately 20 s. Although more detail can be captured using point-cloud-based segmentation, the point-cloud-based segmentations are constrained by computing efficiency, especially for large-scale applications. Considering that the efficiency of the region-merging algorithm mainly depends on the number of regions to be merged, a parallel computing of point cloud tiles can greatly improve the efficiency of the algorithm.

Beyond developing a powerful segmentation method suitable for the current study, our study proposed a novel and flexible segmentation framework that could also be easily extended in future works with appropriate scaling of the distance measurement. Furthermore, the results show that mixed forests are relatively difficult to segment. A multiscale segmentation method that uses various stopping thresholds would be another change to address large variations in crown size in mixed forests. In addition, the region-merging method could be used as a worthy complementary or post-processing method to reduce the over-segmentation issue in other individual tree segmentation algorithms.

VI. Conclusion

The successful detection and delineation of individual trees are critical steps in forestry remote sensing applications. This study proposed a novel hierarchical region-merging method for segmenting individual trees from LiDAR point clouds. Experiments were conducted under different forest stands, and the proposed method was compared with two traditional methods (marker-control watershed segmentation and point clouds region-growing method), which revealed that the region-merging algorithm obtained comparably good performance in low-density coniferous and showed improved detection accuracy in forests with high stand density. Furthermore, the proposed method using the global optimal segmentation strategy instead of detecting treetops or valleys between crowns exhibited advantages in the detection of suppressed trees, and the average detection rate improved by more than 10% compared with the other methods. In the algorithm, the ST was relatively sensitive compared with other parameters and should be tuned as a priority for future applications. This bottom–up hierarchical segmentation method could record the whole merging process and served as a flexible framework

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for individual tree delineation, which could provide a basis for algorithm development and forestry application. Further improvements to the region-merging method could be achieved by developing an adaptive stopping criterion or extending it as multiscale segmentation.

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